

Diffusivity: Einstein's relⁿ: $D = \lim_{t \rightarrow \infty} \frac{1}{6} \frac{d}{dt} \langle (\Delta x(t))^2 \rangle$; $\Delta x(t) = x(t) - x(0)$

The derivations would be for the general case, i.e. $x(0) \neq 0$

2 ways to prove:

① Langevin Dynamics

② Fokker-Planck eqⁿ

① Langevin Dynamics Approach:

$\langle \dots \rangle$: Ensemble Average; average over different random kicks. We can assume that an average over all particles can be a proxy to it (well, this is a computational thing, not related to derivation, but related to the work.)

for any particle: net force = drag force + Random Brownian force

$$F_{\text{net}} = m \underline{a} = -\gamma \underline{v} + \underline{\eta}(t)$$

$$m \frac{d\underline{v}}{dt} = -\gamma \underline{v} + \underline{\eta}(t)$$

$\downarrow \cdot \underline{n}$

$$m \underline{n} \cdot \frac{d\underline{v}}{dt} = -\gamma \underline{x} \cdot \underline{v} + \underline{x} \cdot \underline{\eta}$$

$$\underline{n} \cdot \frac{d\underline{v}}{dt} = \frac{d}{dt} (\underline{x} \cdot \underline{v}) - \underline{v} \cdot \underline{v}$$

$$\Rightarrow m \frac{d}{dt} (\underline{x} \cdot \underline{v}) - m \underline{v} \cdot \underline{v} = -\gamma \underline{x} \cdot \underline{v} + \underline{x} \cdot \underline{\eta}$$

$\downarrow$ Ensemble average

$$m \frac{d}{dt} \langle \underline{x} \cdot \underline{v} \rangle - m \langle \underline{v} \cdot \underline{v} \rangle = -\gamma \langle \underline{x} \cdot \underline{v} \rangle + \langle \underline{x} \cdot \underline{\eta} \rangle$$

η : random Brownian force; uncorrelated to x

$$\frac{1}{2} m \langle \underline{v} \cdot \underline{v} \rangle = \frac{3}{2} k_B T$$

$$\Rightarrow m \frac{d}{dt} \langle \underline{x} \cdot \underline{v} \rangle + \gamma \langle \underline{x} \cdot \underline{v} \rangle = 3 k_B T$$

$$\text{Take } \langle x, v \rangle = y$$

-(1)

$$\Rightarrow m\dot{y} + \gamma y = 3k_B T$$

$$\Rightarrow \dot{y} + \frac{\gamma}{m} y = \frac{3k_B T}{m}$$

$$y \times \exp\left(\frac{\gamma t}{m}\right) = \frac{3k_B T}{m} \frac{\exp\left(\frac{\gamma t}{m}\right)}{\gamma/m} + C$$

$$\Rightarrow y \exp\left(\frac{\gamma t}{m}\right) = \frac{3k_B T}{\gamma} \exp\left(\frac{\gamma t}{m}\right) + C$$

$$y(t=0) = y_0 \Rightarrow y_0 = \frac{3k_B T}{\gamma} + C$$

$$\Rightarrow y \exp\left(\frac{\gamma t}{m}\right) - y_0 = \frac{3k_B T}{\gamma} \left[\exp\left(\frac{\gamma t}{m}\right) - 1 \right]$$

$$\Rightarrow y = y_0 \exp\left(-\frac{\gamma t}{m}\right) + \frac{3k_B T}{\gamma} \left[1 - \exp\left(-\frac{\gamma t}{m}\right) \right]$$

$$\text{for } t \rightarrow \infty; y = \frac{3k_B T}{\gamma} \quad \text{-(2)}$$

Aside: We can get eqⁿ (2), straight from eqⁿ (1).

$$\langle x, v \rangle: x \neq \underline{r}(t) - \underline{r}(0) = \int_0^t \underline{v}(s) ds$$

$$\Rightarrow \underline{r}(t) = \underline{r}(0) + \int_0^t \underline{v}(s) ds$$

$$\langle \underline{r}(t), \underline{v}(t) \rangle = \langle \underline{r}(0), \underline{v}(t) \rangle + \left\langle \int_0^t \underline{v}(s) ds, \underline{v}(t) \right\rangle$$

Assume $\underline{r}(0) = 0$

$$\Rightarrow \langle \underline{r}(t), \underline{v}(t) \rangle = \int_0^t \langle \underline{v}(t), \underline{v}(s) \rangle ds$$

Now, for a large value of t , $\underline{v}(t)$ is uncorrelated w $\underline{v}(s)$
(Vel. Aut. $\rightarrow 0$)

$$\Rightarrow \langle \underline{r}(t), \underline{v}(t) \rangle \rightarrow \text{a constant value, i.e. } \dot{y} \rightarrow 0 \text{ for large } t$$

$$\Rightarrow y = 3k_B T / \gamma$$

for $t \rightarrow \infty$; $\langle \underline{r} \cdot \underline{v} \rangle = 3 k_B T / \gamma$

$$\frac{d}{dt} \langle r^2 \rangle = 2 \frac{d}{dt} \langle \underline{r} \cdot \underline{v} \rangle$$

$$\Rightarrow \frac{1}{2} \frac{d}{dt} \langle r^2 \rangle = \langle \underline{r} \cdot \underline{v} \rangle = \frac{3 k_B T}{\gamma}$$

$$\Rightarrow \frac{1}{6} \frac{d}{dt} \langle r^2(t) \rangle = \frac{k_B T}{\gamma} \quad : \text{for 3-dimensions; } k_B T / \gamma = D$$

for $t \rightarrow \infty \Rightarrow D = \frac{1}{6} \frac{d}{dt} \langle r^2(t) \rangle$

If we use $\underline{r}(0) \neq 0$ & $\underline{v}(t) = d/dt \Delta \underline{r}(t)$;

we'll have $D = \frac{1}{6} \lim_{t \rightarrow \infty} \langle (\Delta \underline{r}(t))^2 \rangle$

Now; $\underline{v}(t_1), \underline{v}(t_2)$: Stationary (i.e. it's $f(t_1 - t_2)$)

but $\underline{r}(t_1), \underline{r}(t_2)$: Not a stationary qty; but still we use (for diff t values) $\langle \underline{r}(t_0), \underline{r}(t_0 + \tau) \rangle$. Why?

well, $\Delta \underline{r} \cdot \Delta \underline{r}$ behaves approximately as a stationary variable for long time

This is why in computations; we leave some equilibration time as well.

(2) Fokker-Planck eqⁿ: @ any position $\underline{r}$ (it's a point in space, not the position of a particle, so $\partial^2 f / \partial x^2 = 0$) and t , PDF = $f(\underline{r}, t)$

$$\frac{\partial f}{\partial t} = D \nabla^2 f$$

$$\langle (\Delta \underline{r})^2 \rangle = \int (\Delta \underline{r})^2 f(\underline{r}, t) d\underline{r}$$

Take the time derivative:

$$\frac{d}{dt} \langle \Delta r^2 \rangle = \frac{d}{dt} \int_{\text{space}} (\Delta r^2) \rho(\underline{r}, t) d\underline{r}$$

$$\frac{d}{dt} \langle \Delta r^2 \rangle = \int_{\text{space}} (\Delta r^2) \frac{\partial \rho}{\partial t} d\underline{r}$$

$$\frac{d}{dt} \langle \Delta r^2 \rangle = \int_{\text{space}} (\Delta r^2) \partial \nabla^2 \rho d\underline{r}$$

$$= \int_{\text{space}} (\Delta r^2) \partial \nabla \cdot (\nabla \rho) d\underline{r} = \partial \int_{\text{space}} \Delta r^2 \nabla \cdot (\nabla \rho)$$

$$\nabla \cdot (a \underline{b}) = (\partial_i a) b_i + a \partial_i b_i$$

$$\Rightarrow \frac{d}{dt} \langle \Delta r^2 \rangle = \partial \int_{\text{space}} (\nabla \cdot (\Delta r^2 \nabla \rho) - \nabla (\Delta r^2) \cdot \nabla \rho) d\underline{r}$$

Now, as $|\underline{r}|$ or $|\underline{r}'| \rightarrow \infty$; ρ or $\nabla \rho \rightarrow 0$ (or $\underline{0}$)

Using div. thm:

$$\frac{d}{dt} \langle \Delta r^2 \rangle = \partial \left[\int_{\text{space}} \Delta r^2 \nabla \rho \cdot \underline{n} ds - \int_{\text{space}} \nabla (\Delta r^2) \cdot \nabla \rho d\underline{r} \right]$$

$$\frac{d}{dt} \langle \Delta r^2 \rangle = -\partial \int_{\text{space}} (2 \Delta r \cdot \underline{\nabla} r) \cdot \nabla \rho d\underline{r}$$

$$\Rightarrow \frac{d}{dt} \langle \Delta r^2 \rangle = -\partial \int_{\text{space}} 2 \Delta r \cdot \nabla \rho d\underline{r}$$

$$\frac{d}{dt} \langle \Delta r^2 \rangle = -2 \partial \int_{\text{space}} \Delta r \cdot \nabla \rho d\underline{r}, \quad \Delta r \cdot \nabla \rho = \nabla \cdot (\rho \Delta r) - \rho \nabla \cdot \Delta r$$

$$\Rightarrow \frac{d}{dt} \langle \Delta r^2 \rangle = -2 \partial \int_{\text{space}} (\nabla \cdot (\rho \Delta r) - \rho \nabla \cdot \Delta r) d\underline{r}$$

$$= -2 \partial \left[\int_{A_0} \rho \Delta r \cdot \underline{n} ds - \int_{\text{space}} \nabla \cdot \Delta r \rho d\underline{r} \right], \quad \nabla \cdot \Delta r = \nabla \cdot \underline{r} = 3$$

$$\int \rho d\underline{r} = 1$$

$$\Rightarrow \frac{d}{dt} \langle \Delta r^2 \rangle = 6D$$

$$\Rightarrow D = \frac{1}{6} \frac{d}{dt} \langle (\Delta r(t))^2 \rangle$$

Where is the $t \rightarrow \infty$ assumption hiding?

Well, to write the eq: $\partial \rho / \partial t = \partial \nabla^2 \rho$

We're to be in the diffusive regime, i.e.

$t \gg \tau_{\text{characteristic relaxation time}} \sim t \rightarrow \infty$